

One-Way ANOVA — Part A

Introduction to One-Way ANOVA Student Worksheet

Name: _____

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Scenario: A professor in the College of Science and Mathematics at Cal Poly SLO wants to understand whether the **study method** a student uses affects their performance on a unit exam in introductory statistics. She randomly assigns 30 students to one of three study conditions: **(A)** traditional re-reading of notes, **(B)** active recall with flashcards, **(C)** AI-assisted study with structured prompting. After the unit, all students take the same 100-point exam. The professor records each score and wants to determine whether differences in mean performance across groups could be explained by random variation alone, or whether there is real evidence of an effect.

1 Research Question

State the research question for this study in a complete sentence. Your question should identify what is being compared and for whom. Write the research question.

Are the population mean of exam scores the same across the different students methods (re-reading (A), active recall (B), AI-Assisted (C))?

2 Variables

(a) State the variables

List each variable mentioned or implied in the scenario.

Variable Name	Description
Study Methods	A condition that is re-reading (A), active recall (B), or AI Assisted (C)
<u>Exam Score</u>	Score on the unit exam, measured 0-100 points

(b) Identify the Explanatory and Response Variables

Role	Variable (name it)
Explanatory (EV)	Study Method (A, B, C)
Response (RV)	<u>Exam Score (0-100 points)</u>

(c) Type of Each Variable

Variable	Type	How many categories / units?
Study Method (A, B, or C)	Categorical	3 Levels: A, B, C
Exam Score (0-100 points)	Quantitative	Points (0-100)

3 Hypotheses

State the null and alternative hypotheses using proper notation. Let μ_A , μ_B , μ_C denote the population mean exam score for each study method group.

Null Hypothesis (H_0)

The population mean of exam score are the same across all three study methods (A, B, C)

$$\mu_A = \mu_B = \mu_C$$

Alternative Hypothesis (H_a)

At least one population of exam scores is not equal to the remaining population mean exam score

Caution: The alternative hypothesis for ANOVA does not specify **which** group differs — only that **at least one** mean is different. A significant result requires follow-up (post-hoc) tests to identify which pairs differ.

4 Analysis

The ANOVA output table below summarizes results from this study. Use it to answer the questions that follow.

Source	df	SS	MS	F (TS)	p-value
Between Groups	2	428.5	214.3	9.14	0.003
Within Groups (Error)	27	633.8	23.5		
Total	29	1062.3			

(a) Identify the test statistic

The test statistic for a one-way ANOVA is the **F-statistic**. From the table:

$F = 9.14$ This means between-group variability is approximately 9 times larger than within-group variability.\

(b) State and interpret the p-value

$p\text{-value} = 0.003$

The probability of observing the sample data or extreme given the null hypothesis true. Based on this p-value we would reject the claim (null hypothesis) that the population means of exam score are the same across study methods (A, B, C).

(c) Write a conclusion in context. Write a conclusion that addresses the research question.

The data provides strong evidence that at least one of the population means of exam scores regarding study methods are different from the others ($F=9.14$, $p = 0.003$). Because students were randomly assigned to groups, this result supports causal claim that study method affected exam performance.